Total No. of Questions : 8]

SEAT No. :

PB2531

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B.E. (Computer Engineering) (Honors in IoT)
MACHINE LEARNING FOR INTERNET OF THINGS
(2019 Pattern) (Semester-VII) (410601)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Explain CNN with any one application. **[6]**

b) Explain Random forest algorithm with working. **[6]**

c) What are the steps to apply PCA in computer vision? **[6]**

OR

Q2) Write short note on following **[18]**

a) Recurrent Neural Network

b) Principle Component Analysis

c) Decision Trees

Q3) a) Explain Hardware implementation of Least-Squares Solvers for Shallow Neural Network. **[6]**

b) What is the need Machine learning accelerators? List and explain different ML accelerators. **[6]**

c) What is edge computing on IoT devices? What are the benifits of it. **[5]**

OR

Q4) a) What is Edge Computing? Why are edge devices essential for IoT? **[6]**

b) Explain Machine Learning Accelerator with an example. **[6]**

c) Explain the concept of Hardware Implementation in Machine Learning. **[5]**

P.T.O.

Q5) a) How deep learning is used in IoT? Explain with example. **[6]**

b) What is embedded deep learning? Explain in brief with benefits of it. **[6]**

c) Explain various deep learning models for sensor data. **[6]**

OR

Q6) a) Explain any one application for deep learning for sensor data. **[9]**

b) For Pre - training the network which deep learning architecture can be used? Why? **[9]**

Q7) a) How IoT is useful in Agriculture? Explain with example. **[6]**

b) Enlist applications of IoT. Explain any two applications in detail. **[6]**

c) How machine learning is useful in IoT security? Explain with example. **[5]**

OR

Q8) a) What is Remote Patient Monitoring? Explain with example. **[9]**

b) Explain in detail Smart Transportation System. **[8]**